

LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

Authoritative doc for the current TX work direction.

[!] The older EXECUTION_PLAN.md (Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).

Branch: tx-rebuild (the direct-port work lives here; origin/main carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

Reference tree (the ONLY trusted TX reference):

- D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\
 - wdsp*.c — WDSP DSP engine (GPL v3+, port-with-attribution OK)
 - ChannelMaster*.c — wire layer (GPL v3+, port-with-attribution OK)
 - Console*.cs — C# UI/control glue (study ONLY; write Lyra-native equivalents)

- Y:\Claude local_hl2src\ — HL2+ ak4951v4 gateway RTL

(control.v / radio.v / usopenhpsdr1.v etc. — the ground truth when host-side reference and gateway disagree)

Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: **"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."** Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing AAMix.{h, cpp}, CMaster.cpp, RadioNet.cpp etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (Reference: Thetis ChannelMaster cmaster.c:297-313, // Per openHPSDR networkproto1.c:1078, // HL2+ ak4951v4 gateway control.v:209-220). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like **"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"** is fine — it matches the public GitHub posture and makes the history grep-able. The earlier 4cef88d tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. ****Public docs (README, docs/*.md design docs, memory files, this plan) — OPEN ATTRIBUTION.**** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. **"AAMix output (HL2 jack)"** not **"Thetis-porting AAMix output"**. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...) + the HL2+ ak4951v4 gateway RTL (Y:\Claude local_hl2src\). **Old**

Python Lyra is NOT a trusted TX reference (its TX never worked).

Consequence of the 2026-06-09 amendment: the three historical commits (efdde02, 604d87b, ed9f0cd) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETE**d by this amendment.

LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION_LOG.md for the full trail.

1. ****Smallest revertable empirical step -> operator HL2 bench ->**

next step.** Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked

2026-05-30 after the fexchange0 crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. ****Read the reference FIRST, in shipped-code form, before**

designing.** Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** git fetch origin is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** _backups/lyra-cpp-YYYY-MM-DD-.bundle.

git bundle create ... --all (covers all refs).

STATUS REGISTRY — what's done, in-progress, pending

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

- [DONE] DONE + operator-bench-validated
- [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)
- [PEND] PENDING (planned, not started)
- [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

Wire layer (ChannelMaster — port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
WDSP exports linkage (OpenChannel/fexchange0/analyzers/PS family)	src/wire/wdspcalls.{h,cpp} (P0.a)	[DONE]	P0.a e0c0f4a 2026-06-09 . The single operator-approved linkage seam: fn_ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + flush_resample added at P0.c.
ChannelMaster\cmcomm.h (family common header)	src/wire/cmcomm.{h,cpp} (P0.b -> P0.d)	[DONE]	complex typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/EER/VOX/TXGAIN/ANALYZERS) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under #pragma once).
ChannelMaster\ilv.c (TX I/Q interleaver)	src/wire/ILV.{h,cpp} (P0.b)	[DONE]	P0.b 1f49d98 2026-06-11 . VERBATIM direct port (ilv, *ILV twin typedef, raw Outbound fn_ptr, pcm->xmtr[] .pilv setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier src/wdsp/ILV retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.
ChannelMaster\cmbuffs.{h,c} (CMB ring + cm_main pump)	src/wire/CmBuffs.{h,cpp} (P0.d)	[DONE]	P0.d afc7950 2026-06-12 . VERBATIM rewrite (cmb, *CMB, #define CMB_MULT (3), malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; pcm->in[] alloc moved back to create_cmastest per reference). Mechanical diff = whitespace-only. [DONE] P0.d operator HL2 RX-regression bench PASSED 2026-06-12 .
ChannelMaster\cmsetup.{h,c} (radio structure + id helpers)	src/wire/cmsetup.{h,cpp} (P0.d, NEW)	[DONE]	P0.d afc7950. VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,...) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\cmaster.{h,c} (FULL: struct + create/destroy + xcmaster + setters)	src/wire/CMaster.{h,cpp} (P0.d)	[DONE]	P0.d afc7950. VERBATIM rewrite: full _cmaster struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), cmaster, *CMaster, raw TCI callback fn ptrs, enum AudioCODEC, cm = {0}. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — the TxChannel RAIL carve-out is DELETED , src/wdsp/TxChannel.{h,cpp} retired). create_cmaster/destroy_cmaster verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-voix/txgain/peer/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags. Sole code accommodation: (char*)" at XCreateAnalyzer. [DONE] Operator HL2 RX-regression bench PASSED 2026-06-12 (RX working on the new per-stream pump/ring layout).
ChannelMaster\obbuffs.c (TX-out ring/seam)	src/wire/ObBufs.{h,cpp} (P1, NEW)	[WIP]	P1 SHIPPED 2026-06-12 — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). Sole deferred line: sendOutbound(id, a->out) in ob_main (network.c:1237 = P2 scope), carried as reference text. TU is DORMANT — no Lyra caller of create_obbuffs until P3/P4; wire bytes unchanged.
ChannelMaster\networkproto1.c (HL2 wire writer)	src/hl2_stream.cpp (LIVE) + src/wire/* (Step 14 pieces)	[WIP]	This is the file the Step 14 plan was attacking. Currently a MIX of Step 14 Lyra-native rewrites + the original hl2_stream.cpp body. P2 = sendOutbound/sendProtocol1Samples fidelity audit of the dormant wire layer; P4 = Wire-LIVE switchover (one commit, full HL2 bench gate).
ChannelMaster\network.c (UDP + keepalive)	src/wire/Ep2SendThread.{h,cpp} + src/wire/Ep6RecvThread.{h,cpp}	[WIP]	Step 14 Lyra-native rewrites shipped (\$5/\$6); could be re-reported direct from reference, OR left as-is if reference-parity audit passes. Operator call (folds into P2/P4).
ChannelMaster\netInterface.c: :create_rnet	src/wire/RadioNet.cpp::create_rnet (LIVE)	[DONE]	Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit 8b8e0da) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering).

WDSP RX chain (port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\amix.{c,h}	src/wire/AAMix.{h,cpp} (P0.c)	[DONE]	P0.c e0f2584 2026-06-11 — VERBATIM re-port (supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class). amix, *AAMIX twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs amix.{h,c} = whitespace-only. src/wire/resample.h = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE] Operator HL2 RX bench PASSED 2026-06-11.
wdsp\firmin.c (filter cores)	(consumed via WDSP cffi)	[N/A]	Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via _native/ per the Python-project pattern; cdef + dlsym in wdsp_native.cpp). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.
wdsp\wisdom.c	(consumed via WDSP cffi)	[N/A]	Same as firmin.c.

WDSP TX chain (per CLAUDE.md §4.1 port table)

Source	Target (lyra-cpp)	Status	Notes
wdsp\TXA.c (channel scaffold)	(consumed via wire/wdspcalls seam — P0.a/P0.d)	[DONE]	TxChannel class DELETED at P0.d afc7950 — the verbatim create_xmtr in wire/CMaster.cpp opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAI justification the class existed for. wdsp_native.{h,cpp} still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface.
wdsp\bandpass.c (bp0/bp1)	(consumed via WDSP cffi — SetTXABandpassFreqs)	[N/A]	The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.

Source	Target (lyra-cpp)	Status	Notes
wdsp\compress.c (TX compressor)	src/wdsp/Compress.{h,cpp} OR cffi binding	[PEND]	Operator choice: direct port (~150 LOC) for v0.2.1 EQ + dynamics ship, OR cffi-only binding if WDSP DLL exports are sufficient. Direct port recommended for parity with Lyra's existing UI control surface.
wdsp\cfcomp.c (5-band CFC)	src/wdsp/CFComp.{h,cpp} OR cffi binding	[PEND]	Same operator-choice. ~600 LOC; one of the heavier ports. Replaced in Python Lyra plan by a Lyra-native Combinator per §15.19; cpp may take the same path (custom Combinator instead of WDSP CFC) — operator design call.
wdsp\osctrl.c (CESSB)	cffi binding (SetTXAosctrlRun)	[PEND]	Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.
wdsp\wcpagc.c mode 5 (leveler)	cffi binding (SetTXALeveler*)	[WIP]	cdef already present in wdsp_native.cpp. UI surface exists in TX-1 components on main. Confirm reference-parity at the call-site setter sequence.
wdsp\wcpagc.c mode 5 (ALC)	cffi binding (SetTXAALCMaxGain)	[WIP]	cdef present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling).
wdsp\eqp.c (parametric EQ)	src/wdsp/EQ.{h,cpp} OR cffi binding	[PEND]	Per CLAUDE.md §4.1 v0.2.1; ~300 LOC. Lyra-native parametric EQ likely (§15.19 lists 3-or-5-band parametric replacing the WDSP 10-band graphic).
wdsp\gen.c (gen0/gen1 generators)	cffi binding (SetTXAPreGen*/SetTXAPostGen*)	[WIP]	gen1 (postgen) used today for TUN tone. gen0 (pregen) cdefs deferred per §15.23 (bench-tooling nicety, not on the signal path).

TX PureSignal (per CLAUDE.md §4.1 v0.3)

Source	Target (lyra-cpp)	Status	Notes
wdsp\iqc.{c,h}	src/wdsp/Iqc.{h,cpp} (~315 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).
wdsp\calcc.{c,h}	src/wdsp/Calcc.{h,cpp} (~1164 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI.
wdsp\lmath.c::xbuilder	src/wdsp/PsXBuilder.{h,cpp} (~200 LOC)	[PEND]	v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.
wdsp\delay.c	src/wdsp/DelayLine.{h,cpp} (~80 LOC)	[PEND]	v0.3. TX/feedback time-alignment.

Console (C# — study only, Lyra-native equivalents)

Source	Target (lyra-cpp)	Status	Notes
Console\console.cs::chkMOX_CheckedChanged2	src/ptt.{h,cpp} + src/wdsp_engine.cpp keydown/keyup	[WIP]	Lyra-native FSM shipped on main. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). No direct port — C# is study-only by license.
Console\console.cs (TR delays)	src/tx/TrSequencing.{h,cpp} (LIVE)	[DONE]	Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/space_mox=13/ptt_out=5/key_up=10 ms).
Console\console.cs::m_bATTonTX	src/tx/AttOnTxPolicy.{h,cpp} (LIVE)	[WIP]	Lyra-native shipped on main. Reference posture: keydown save per-band step-att + force-31 (when PS-A off OR CW); keyup restore. Implementation verified against console.cs:30293-30327 + :30391-30410.
Console\PSForm.cs	src/ui/PsDialog.{h,cpp} (NEW, v0.3)	[PENDING]	v0.3 PureSignal UI. C# study only; write Qt/QML native.
Console\HPSDR\IoBoardHL2.cs	src/hl2_stream.cpp (LIVE)	[WIP]	HL2 I/O quirks already implemented Lyra-native via the protocol-byte verification + gateway-RTL ground-truth pass (CLAUDE.md §15.26).

STAGE INDEX — ordered work plan

P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-ported file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — lyra:wire namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

Step	Status	What	Commit / gate
P0.a	[DONE] DONE	wire/wdspcalls.{h,cpp} — the single approved WDSP linkage seam (exact export names, PS entry family complete)	e0c0f4a 2026-06-09
P0.b	[DONE] DONE	ilv.{h,c} verbatim -> wire/ILV.{h,cpp} + new wire/cmcomm.{h,cpp}	1f49d98 2026-06-11; test_ilv ALL PASS
P0.c	[DONE] DONE	aamix.{h,c} verbatim -> wire/AAMix.{h,cpp} + wire/resample.h RESAMPLE ABI	e0f2584; [DONE] operator HL2 RX bench PASSED 2026-06-11
P0.d	[DONE] DONE	cmbuffs/cmaster/cmsetup verbatim -> wire/CmBuffs/wire/CMaster/wire/cmsetup; full_cmaster struct (PS surface); TxChannel carve-out DELETED	afc7950 2026-06-12; [DONE] operator HL2 RX-regression bench PASSED 2026-06-12
P1	[WIP] SHIPPED	obbuffs.c verbatim -> wire/ObBuffs.{h,cpp} (the TX-out seam; separate TU from cmbuffs; sendOutbound call DEFERRED to P2)	2026-06-12; dormant/wire-inert — optional operator RX smoke; clean build 0 warnings; mechanical diff IDENTICAL
P2	[PEND]	sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer	

Step	Status	What	Commit / gate
P3	[PEND]	netInterface outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (verbatim setters have NO null guards)	
P4	[PEND]	Wire-LIVE switchover	ONE commit, full HL2 bench gate

PureSignal is a committed feature — the P0.d _cmaster struct carries the full PS surface so v0.3 lands on top without a retrofit.

Historical stage index (pre-P0 lettered stages — superseded)

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-reported verbatim at P0.c.

Stage	Status	What	Notes
A	[DONE] DONE	CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun)	Superseded by the P0.d full verbatim CMaster port.
B	[DONE] DONE	aamix.c direct port + wire into RX path	Shipped 2026-06-08 (idiom-translated); re-reported VERBATIM at P0.c.
C	[DONE] DONE (as P0.b)	ilv.c direct port (TX I/Q interleaver)	Landed verbatim at wire/ILV.{h,cpp}.
D	[DONE] DONE (as P0.d)	xcmaster pump body direct port	Landed verbatim in wire/CMaster.cpp (real stype/txid/chid + update[] CS).
E	[N/A] OBSOLETE	TxChannel reference-parity audit + reconciliation	TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.
F	[PEND] TBD	WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1	compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.
G	[PEND] TBD	PureSignal direct ports (v0.3)	iqc / calcc / xbuilder / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod.
H	[PEND] TBD	TX/PS auto-attenuator FSM port	C# study only (PSForm.cs); Lyra-native rewrite.

Ordering note: the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertable step + mechanical diff + operator bench gate) applies within each step.

OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. **Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E

(TxChannel audit) — which first?** All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).

- Stage E formalizes what's already shipped (lowest risk, highest confidence-gain).

2. ****Do we re-port networkproto1.c + network.c + the Step 14**

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original `hl2_stream.cpp`?** The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. ****WDSP TXA-chain components — direct port (~150-600 LOC**

each) vs cffi-only?** The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source-level need surfaces.

4. ****PS work timing — start v0.3 (iqc/calcc/etc.) now in**

parallel with TX, or sequence after TX is fully done?** The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. ****The §15.27 "opt-in toggles that become defaults" cleanup**

from the Python project — does it have a lyra-cpp analog here?** The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

SHIPPED-WORK LOG

Newest at top. Cross-references SESSION_LOG.md for full arc details.

2026-06-12 — P0.d cmbuffs/cmsetup/master/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

- **P0.d** afc7950 on tx-rebuild: NEW `wire/cmsetup.{h,cpp}`

(`cmMAX* 16/4/4/2/32` + the 8 id helpers + `SetRadioStructure` + `set_cmdefault_rates`); `wire/CmBuffs.{h,cpp}` rewritten verbatim (`cmb, *CMB, malloc0/_aligned_free`; Phase-C retrofit deviations removed); `wire/CMaster.{h,cpp}` rewritten verbatim (FULL `_cmaster` struct incl. the `PureSignal` surface, enum `AudioCODEC`, raw TCI fn ptrs, verbatim `create/destroy_cmsetup` + `create/destroy_xmtr` + `xcmaster` + all `Sendp*/Set*` setters).

- **TxChannel RAIL carve-out DELETED** (`src/wdsp/TxChannel.{h,cpp}`)

retired) — the verbatim `create_xmtr` opens the WDSP TXA channel + TX analyzer (`disp 1`) + `out[0..2]` + ILV itself through the `wdspcalls` seam.

- `main.cpp`: `SetRadioStructure(2,1,1,1,0,...)` config before

`create_cmsetup` (`chid(0,0)=0` RX1 / `chid(1,0)=1` TX = the live layout); `create_xmtr()` post-resolve `wdsp_calls` (documented DEFERRED-CALLSITE accommodation); `handler-1.5` = gated `destroy_xmtr()`.

- Verification: clean build ZERO warnings; `test_ilv` ALL PASS;

mechanical diffs IDENTICAL (`CmBuffs.{h,cpp}` / `cmsetup.h` / `CMaster.h` / all 10 fully-verbatim `cmsetup.c` functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: `(char*)""` at `XCreateAnalyzer`).

- New at startup: TWO `cm_main` pump threads (`streams 0+1`; `stream 0`

`idles` — no `Inbound producer`; TX stays wire-quiescent (`OutboundTx` null until P3).

- [DONE] **GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (`obuffs.c`) is unblocked.

2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

- Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

- **P0.a** e0c0f4a: `wire/wdspcalls.{h,cpp}` — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified).
scratch/_typedef_fidelity_test.cpp disproved the claimed "C++ collisions" (twin typedef / complex[2] / raw fn ptr all compile verbatim).

- **P0.b** 1f49d98: ilv.{h,c} verbatim -> wire/ILV.{h,cpp};

new wire/cmcomm.{h,cpp} (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test_ilv ALL PASS.

- **P0.c** e0f2584: aamix.{h,c} verbatim -> wire/AAMix.{h,cpp}

(supersedes the Stage-B idiom translation); wire/resample.h = public RESAMPLE ABI; wdspcalls retyped + flush_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**

2026-06-08 EOD — Stage B aamix.c port COMPLETE

- **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated

- Commits 27ac2fa -> 7a50b07 on tx-rebuild

- Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

- B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

RadioNet.cpp:272 clobbered aaMix_ -> Outbound ~1 ms after WdspEngine::openRx1 wired the real lambda. Fixed by deleting the stub (commit 8b8e0da).

- 2-stage diagnostic instrumentation (chain @ 225518c +

dispatch @ 12369bc) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

- Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

- Tasks #130 (parent) + #132 (Stage B.6.b) closed.

- Backup _backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE-with-docs.bundle.

Earlier work

See SESSION_LOG.md for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in EXECUTION_PLAN.md's progress- tracking sections. TX-1 components 2-6 (main branch) are covered in project_lyra_cpp_tx.md (memory file).

RESUME POINTER (next session)

First actions (locked discipline):

1. git fetch origin
2. git status + git log --oneline -5 tx-rebuild origin/main
3. Read this doc + the "READ FIRST" block in project_lyra_cpp_tx.md
4. Read SESSION_LOG.md newest entry for arc detail

Then: wait for operator decision on which Stage (C / D / E / F / open-questions item) to land next. Each Stage gets its own grounded design + reference-read + smallest-revertable- step + bench-gated implementation pattern matching the Stage B arc.

Last updated: 2026-06-08 EOD — Stage B aamix.c port shipped + bench-validated. Next port target = operator's call.